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Extracting Commonsense Knowledge for Robotic Agents from LLMs

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Abstract

The acquisition of commonsense knowledge remains a core challenge in both robotics and artificial intelligence. While Large Language Models (LLMs) encode rich latent knowledge, their unstructured and probabilistic nature limits their direct use in safety-critical domains like robotics. In this paper, we present a pipeline for extracting structured commonsense knowledge from LLMs and integrating it into a symbolic knowledge graph based on the Ontology for Robotic Knowledge Acquisition (ORKA). Grounded in the theory of conceptual spaces, our method targets physical object properties—both categorical (e.g., shape, material, location) and quantifiable (e.g., size, weight, temperature)—relevant to embodied reasoning. We evaluate the pipeline across a diverse set of LLMs, model sizes, and quantization levels, using human-annotated ground truth to assess accuracy. Results show that even compact and quantized models can produce reliable, interpretable knowledge. We also analyze the influence of measurement units and contextual relevance, highlighting trade-offs between model types and use cases. These results indicate that even smaller LLMs can serve as a useful intermediate source for deriving structured commonsense representations in robotics contexts.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Commonsense Knowledge, Knowledge Graph Construction, Large Language Models, Robotics

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1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities across a wide range of domains, from natural language understanding to problem solving and robotics. Their ability to encode vast amounts of commonsense knowledge implicitly allows them to generate highly contextual and often surprisingly accurate outputs. For many tasks, this represents a breakthrough in data-driven reasoning. However, the latent and unstructured nature of this knowledge limits suitability for real-world deployment in safety-critical domains such as robotics, where decisions must be interpretable, reliable, and grounded. In embodied settings, probabilistic and opaque behavior introduces operational risks [17].

To address this problem, structured representations such as commonsense knowledge graphs offer a compelling alternative [5, 16, 27]. Unlike LLMs that are black-box by nature, knowledge graphs provide transparent, query-able structures that can encode human-like reasoning over object properties, spatial relations, and contextual knowledge. A range of prior efforts have explored this space using manually curated (e.g., CyC [19]), semi-automated (e.g., DBpedia [23], ATOMIC [14]), and fully automated methods [15] — each facing the trade-off between quality and scalability. Such graphs have proven to be effective in robotics applications, supporting object perception [33], affordance reasoning [32], and high-level planning [29] with clearer semantics and tighter control. This growing need for both adaptability and interpretability has led to increased interest in neuro-symbolic approaches, which aim to integrate the representational flexibility of neural models with the structured reasoning capabilities of symbolic systems.

In this paper, we argue that effective robotic reasoning requires this hybrid approach that combines traditional symbolic methods—such as knowledge representation, reasoning, and planning — with the commonsense knowledge implicitly embedded within LLMs. Our aim is to explore how the latent, unstructured information within LLMs can be systematically extracted and transformed into a structured, symbolic format suitable for use in reasoning and planning tasks. This problem entails identifying which models are best suited for this extraction and investigating the most effective strategies for producing high-quality, usable commonsense knowledge.

To address this problem, we present a pipeline for constructing symbolic commonsense knowledge graphs derived from LLMs, with a specific focus on object categories and their physical properties. Central to our approach is the selection of quality dimensions—such as color, shape, material, weight, size, disposition, and typical location—used to structure the extracted knowledge. These dimensions are motivated by the conceptual spaces theory proposed by Gärdenfors [12], which models concepts as regions within a geometric space defined by semantically meaningful axes. By adopting these dimensions, we enable a representation that is both human-interpretable and grounded in perceptual and functional

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properties relevant to robotic reasoning. From a practical perspective, these dimensions capture information essential for affordance inference, manipulation planning, and scene understanding in embodied agents. For instance, size and weight can directly influence grasping strategies, while material and temperature can inform safety considerations during interaction. Dispositional attributes (e.g., fragility, flexibility) affect how objects should be handled, and typical location supports context-aware reasoning in task execution. By organizing this information in a structured, symbolic format, we bridge the gap between the raw output of LLMs and the actionable knowledge required by autonomous systems. To provide semantic structure and consistency across dimensions, we organize these properties using the Ontology for Robotic Knowledge Acquisition (ORKA) [1], which defines a set of cognitively and functionally grounded dimensions—such as color, shape, material, size, weight, and disposition—relevant for reasoning in embodied agents.

We evaluate our pipeline across multiple LLMs, and the concepts of the MS COCO dataset [21], examining how factors such as model architecture, parameter count, and quantization influence the quality of extracted knowledge[1]. In addition to model-level variation, we also analyze how performance is affected by the type of information extracted—specifically, differences across conceptual dimensions (e.g., size, weight, material) and the presence of varying measurement units (e.g., grams vs. kilograms, centimeters vs. meters). Our results show that even relatively small models ($\sim$ 4B parameters) can produce knowledge that aligns closely with human annotations, highlighting the feasibility of using LLMs as a source of structured commonsense information for downstream symbolic reasoning.

2 Related Work

Large Language Models (LLMs) have recently gained traction as powerful tools for embedding and using commonsense knowledge within robotic systems. This has spurred a wide array of research aimed at integrating LLMs into different components of robotic control architectures. Vision-Language Models (VLMs), for instance, have been explored for their capacity to connect perceptual input with semantic understanding [10, 13]. Beyond perception, LLMs have been employed in planning modules [4, 20], action generation [13], and human-robot interaction [16]. Recent work also explores the role of LLMs as high-level controllers or advisors within robot architectures [25], highlighting their potential to provide commonsense reasoning and flexible task adaptation across a variety of domains. These efforts reflect a broader shift toward leveraging LLMs not merely as language processors but as general-purpose reasoning agents capable of supporting more adaptable and intelligent robotic behavior.

Commonsense Knowledge Graphs. Numerous efforts have aimed to capture structured commonsense knowledge through various methodologies, including expert-driven knowledge engineering (e.g., Cyc [19]), large-scale crowdsourcing (e.g., Wikidata [30], DBpedia [6]), and automated or semi-automated extraction pipelines (e.g., CSKG [15], ATOMIC [14]). These resources have contributed significantly to the field but typically offer limited coverage of commonsense knowledge related to the physical and functional properties of everyday objects, e.g., typical size, weight, material composition, color, and affordances [2]. This gap highlights the ongoing need for structured representations of physical-world knowledge that are both scalable and robust. Our work builds on prior work where affordances are framed in terms of object dispositions under specific enabling conditions [9], and extends the notion of object capabilities introduced in [8] to guide the extraction of affordance-related knowledge from LLMs. The automatic construction of knowledge graphs from web-scale data dates to efforts such as WebChild [28] and ASCENT++ [26], that leveraged semi-supervised and unsupervised learning techniques to extract structured commonsense knowledge. Large Language Models have been recently employed as a source of symbolic knowledge via the process of knowledge distillation, to either expand open-domain knowledge bases (such as ATOMIC in [31],or to improve domain-specific contexts (e.g., the e-commerce in [34]). These studies provide critical evidence that symbolic knowledge can be reliably distilled from the implicit representations within LLMs—an idea that our work extends to the physical and affordance properties of objects relevant to robotics.

Knowledge Graph Construction Using LLMs. Constructing high-quality knowledge graphs typically demands substantial human effort, making automated Knowledge Graph Construction (KGC) using Large Language Models an increasingly attractive research direction. Several recent works have proposed pipelines to reduce manual intervention in this process. For instance, [18] introduce a semi-automatic, LLM-assisted pipeline for end-to-end ontology and knowledge graph construction. As the schema often needs to be tightly integrated with the system's internal representations, our work assumes that the ontology is predefined and grounded within the broader robotic architecture. Other researchers have explored prompting strategies and extraction methods to guide LLMs in generating graph content. [11] investigates various prompting techniques with GPT-3.5 for iterative KGC, [35] focuses on the extraction of relational triples from unstructured text, and [36] proposes a framework for fine-tuning LLMs to produce generalized, domain-agnostic knowledge graphs. An evaluation of GPT-4's ability to perform KGC with previously unseen information is found in [37]. In contrast to these surface-level extraction works, we target the implicit commonsense knowledge embedded within LLMs, aiming to distill this latent knowledge into a symbolic format grounded in object properties and affordances relevant to robotics. While prior work has advanced automated knowledge graph construction, much of it depends on fine-tuned or proprietary models [11, 35, 36], limiting reproducibility. Other approaches focus on schema induction [18, 36], often assuming no existing ontology. Our approach differs by targeting robotics, where predefined ontologies guide the knowledge-intensive tasks. We use both open-source and proprietary LLMs to extract structured knowledge about object properties and affordances—an underexplored area in KGC—aiming for symbolic, grounded representations usable in embodied, safety-critical systems.

[1]Data, instructions for replicability, and additional results are available at https://github.com/Dorteel/commonsense_kg_construction/tree/k-cap-2025

3 Proposed Approach

While commonsense knowledge graphs have been explored extensively, the demands of safety-critical systems—such as autonomous robots—require representations that are both interpretable and grounded in measurable, real-world properties. As outlined earlier, these properties span both categorical dimensions (e.g., color, shape, material, disposition, and typical location) and quantitative dimensions (e.g., size, weight, and temperature), all of which are relevant for robotic perception, manipulation, and planning. However, not all dimensions contribute equally to defining a given concept. For example, when distinguishing between kitchen appliances, dimensions such as size, shape, and typical location are often critical, whereas properties like color or temperature may carry little definitional weight. To model this variability, we adopt the concept of context vectors introduced by Adams et al. [3], which extends the theory of conceptual spaces by characterizing concepts in terms of their most salient dimensions. These vectors guide the selection of relevant attributes during knowledge extraction, enabling the construction of representations that reflect concept-specific structure rather than applying a uniform schema to all categories.

Our methodology integrates these ideas with the Ontology for Robotic Knowledge Acquisition (ORKA)[1] to provide a structure for the proposed pipeline for extracting structured knowledge from language models, focusing on dimensions that are most likely to support decision-making in embodied agents. For quantitative dimensions, we go beyond previous work on relative comparisons [7] by extracting measurement-specific estimates (e.g., average width in centimeters, typical weight in kilograms). These values are intended to support precise spatial reasoning, constraint satisfaction, and motion planning in robotic systems.

3.1 Knowledge Extraction Pipeline

We extract structured commonsense knowledge about everyday concepts using concept labels and descriptions, along with a set of predefined quality dimensions to construct prompts for the LLMs. The output of each model is checked for syntactic and semantic integrity, then parsed through the ontology to ensure grounding and to update the knowledge graph accordingly. This section details each of these steps (see Figure 1 for a high-level overview).

Inputs. For each concept, we start with a label defining the concept and a disambiguated textual description—sourced from the knowledge graph (via `rdfs:comment`)—to provide context and reduce ambiguity. This input is combined with the quality dimensions we aim to query. As mentioned above, the currently considered quality dimensions include the categorical dimensions of material, shape, location, and disposition, and the quantifiable dimensions of weight, temperature, and size.

Prompt Generation. Prompts are dynamically constructed by combining the inputs described above with templates tailored separately for categorical and quantifiable dimensions, since the latter require measurement units. The templates are designed to encourage the models to return responses in a structured JSON format.

Model Selection. The model selection component chooses from a predefined set of available language models. This selection process supports both API-based models and locally deployed models, with

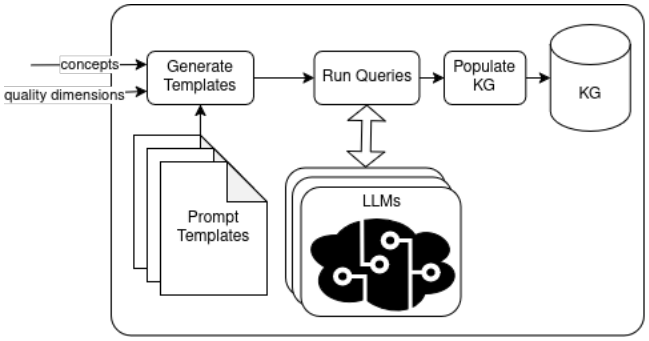

Figure 1: High-level overview of our extraction pipeline.

the latter typically being quantized to accommodate the resource constraints usually present with robotic agents. The choice of model may depend on factors such as model accuracy, latency requirements, and hardware availability. By supporting a hybrid model pool, the pipeline maintains flexibility in deployment environments—ranging from lightweight, on-device use to high-capacity cloud-based inference—while also enabling comparative analysis across different model types and configurations.

Syntax Parsing. Many recent language models support structured output formats, such as JSON or constrained generation using context-free grammars. When available, these features are incorporated to simplify parsing. To accommodate models without native structured output, we rely on prompt engineering and simple heuristics to extract the required information.

Semantic Parsing. Semantic parsing refers here to linking the model outputs to existing concepts in the ontology. For quantifiable dimensions, this involves validating the dimension label and ensuring the output matches the expected data type (e.g., numerical values with appropriate units). For categorical dimensions, we use both alternative labels (e.g., synonyms, variant spellings) and the class hierarchy to resolve references. This helps disambiguate between similar or related terms and ensures consistency when incorporating new data into the knowledge graph.

This pipeline provides a straightforward method for extracting and structuring commonsense knowledge from LLMs, with the output aligned with an existing ontology. With the extracted information integrated into the knowledge graph, we next evaluate its quality and reliability across different models, prompting strategies, dimensions, and units of measurement.

4 Evaluation

To assess the effectiveness of our knowledge extraction pipeline, we conduct a series of experiments aimed at answering the following research questions:

- ***RQ1: Model Performance:*** *Which language model—considering different sizes and quantization levels—produces the most accurate commonsense knowledge for each quality dimension?* We analyze output quality across various LLMs, comparing instruction-tuned versus standard models and covering

different parameter ranges, aiming to help identify optimal configurations for practical use in robotic systems.

- ***RQ2: Measurement Sensitivity:*** *How does the choice of measurement unit (e.g., grams vs. kilograms) affect the quality and consistency of extracted quantitative information?* Since robotic systems require knowledge grounded in specific units (e.g., centimeters, meters, kilograms), **RQ2** evaluates how the choice of unit impacts the precision of LLM outputs.
- ***RQ3: Dimension Relevance:*** *How can the relevance of different quality dimensions be inferred from the variability in model responses? In particular, will less defining dimensions (e.g., temperature) exhibit greater response variation compared to core attributes like weight or size?* We aim to examine the degree of agreement across models of different architectures and sizes regarding the contextual relevance of specific dimensions. We use output variance as a proxy, focusing on *temperature* (for quantifiable dimensions) and *disposition* (for categorical dimensions) as examples of generally non-defining properties. We expect these to exhibit higher variability compared to more central dimensions such as *size* and *color*, respectively.

4.1 Experimental Setup

Our evaluation is based on a set of 80 object concepts drawn from the MS COCO dataset, chosen for their diversity and relevance in everyday contexts. Each concept was parsed through WordNet [24] to obtain a disambiguated definition and to place it within the hierarchical structure of the ORKA ontology. For each concept, we then query the selected language models using the full set of predefined quality dimensions—both categorical (color, shape, material, location, and disposition) and quantifiable (weight, temperature, and size). For quantifiable quality dimensions, each model is queried for the selected measurement units that can correspond to the given dimension, considering both metric and imperial measurements for sizes (cm, mm, in, ft) and weights (g, kg, oz, lb), and three units for temperature ($°K$, $°F$ and $°C$). Each model is queried 20 times for each (concept-dimension-unit) combination to examine statistical significance and consistency of model outputs.

4.1.1 Ground Truth. Since information along these quality dimensions is often sparse or inconsistent across existing sources, we established a ground truth through human annotation. Using Amazon Mechanical Turk, we collected 10 annotations per concept for each quality dimension. Annotators were free to select the unit of measurement for the quantitative dimensions, and these measurements were transformed to the other units automatically using standard conversion rates. Annotators were also asked to indicate per concept which dimensions they considered irrelevant, allowing us to capture both value estimates and contextual relevance.

4.1.2 Models Used. The selection of models used in the experiment was guided by available computational resources and access constraints, while also aiming for diversity in terms of model size, architecture, and developer origin. We included both open-weight and API-based models[2] to reflect practical deployment scenarios and to evaluate trade-offs between accessibility, performance, and cost. Model sizes ranged from approximately 1.5 billion to over 70 billion parameters, allowing us to compare performance across different capability scales. We also included multiple versions of the same model family (e.g., LLaMA 2 vs. LLaMA 3) to observe the impact of architectural updates. Both quantized and full-precision variants were considered, particularly for local deployment scenarios where memory constraints are significant. A complete summary of all models included in the evaluation can be found in Table 1.

[2]https://groq.com/ and https://nebula.cs.vu.nl/

4.1.3 Sources of error. We categorize the observed errors into three main types: syntactic, semantic, and factual. *Syntactic Errors* occur in the syntax parsing module, when the model outputs text that violates the expected structural format. In our case, this typically involves invalid or malformed free-form JSON responses that cannot be parsed. *Semantic Errors* involve structurally correct outputs that nonetheless fail to map meaningfully to the target knowledge graph schema. This includes cases where object attributes or relationships are described in a vague or misaligned manner, rendering them un-parsable for structured integration. These errors highlight the challenge of ensuring conceptual compatibility between LLM output and symbolic knowledge representations. Lastly, *Factual Errors* refer to instances where the content of the model's output is simply incorrect or nonsensical—e.g., stating that an average toaster is typically one meter tall, or that scissors have a disposition of "riding." These mistakes reflect the probabilistic and ungrounded nature of LLMs, especially when generating domain-specific physical knowledge without explicit context or constraints. Factual errors are calculated separately for categorical and quantifiable dimensions as explained below.

4.2 Results

4.2.1 RQ1: Prediction accuracy. To compare the overall performance of the different models, two scores are calculated for each model m. Where the dimensions are quantifiable, the score s_m^{num} is based on the inverse error calculated between the mean predicted value for the given model $\overline{x}^m$ and the mean ground truth $\overline{g}$ and combined across the quantitative dimensions D_{num} and concepts C (Equation 1). For categorical dimensions, the Jaccard similarity index s_m^{cat} is calculated to measure the overlap between the set of categories predicted by the model P^m and the ground truth set G (Equation 2), where both sets were determined using majority voting over the samples. A summary of the results of the experiment as well as the details of the models used can be seen in Table 1 and Figure 2.

$$s_m^{\text{num}} = \frac{1}{C}\sum_{c=1}^{C}\left(\frac{1}{D_{\text{num}}}\sum_{d=1}^{D_{\text{num}}}\frac{1}{1+\left|\overline{x}_{c,d}^m - \overline{g}_{c,d}\right|}\right) \tag{1}$$

$$s_m^{\text{cat}} = \frac{1}{C}\sum_{c=1}^{C}\left(\frac{1}{D_{\text{cat}}}\sum_{d=1}^{D_{\text{cat}}}\frac{\left|P_{c,d}^m \cap G_{c,d}\right|}{\left|P_{c,d}^m \cup G_{c,d}\right|}\right) \tag{2}$$

Overall performance. Results show notable variation across models in both syntactic correctness and semantic accuracy. GPT-4.1 achieves the highest overall numerical performance, with the best score in quantifiable dimensions ($s_{\text{num}} = 0.594$) and the lowest semantic error ($E_{\text{sem}} = 2.27$). In the categorical domain, LLama4 Scout

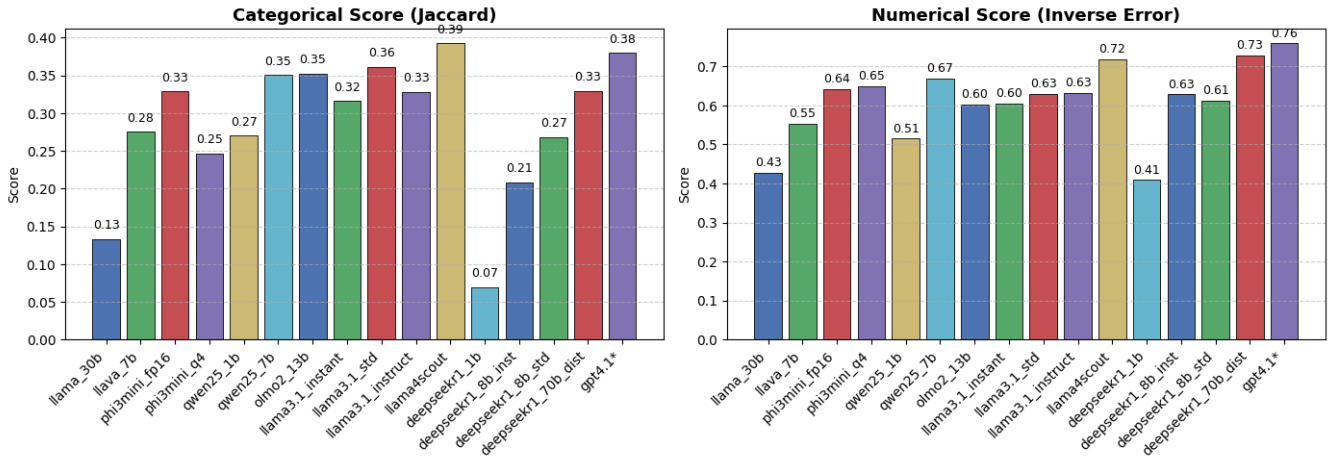

Figure 2: Model scores for the different dimensions, combined across concepts.

Table 1: Model comparison across syntactical and semantic errors, performance scores and error variation for categorical and numerical dimensions as well the percentages of exact matches for categorical domains and mean relative error.

Model	Version	Params.	Developer	E_{syn}	E_{sem}	s_{cat}	s_{exact}	s_{num}	E_{MRE}	σ_{MRE}
LLama	Standard	30b	Meta	0.73	13.99	0.1330	0.048	0.220	12.853	52.127
LLava	Standard	7b	Meta	1.42	7.02	0.2750	0.083	0.342	7.328	47.088
Phi3 mini (fp16)	Instruct	3.8b	Microsoft	**0.01**	4.80	0.3288	0.065	0.444	1.345	5.561
Phi3 mini (q4)	Instruct	3.8b	Microsoft	**0.01**	7.76	0.2468	0.043	0.451	**1.305**	6.130
Qwen2.5	Standard	1b	Alibaba Cloud	0.35	5.33	0.2702	0.053	0.309	7.003	40.323
Qwen2.5	Standard	7b	Alibaba Cloud	0.02	5.72	0.3504	0.078	0.512	1.500	8.037
Olmo2	Instruct	13b	AllenAI	0.02	13.83	0.3524	0.085	0.417	5.011	39.197
LLama3.1	Instruct	8b	Meta	0.08	6.18	0.3159	0.058	0.419	2.219	10.505
LLama3.1	Standard	8b	Meta	1.69	5.56	0.3608	**0.090**	0.411	2.369	10.368
LLama3.1	Instant	8b	Meta	1.21	6.57	0.3283	0.055	0.388	4.610	24.555
LLama4 Scout	Instruct	17b	Meta	**0.01**	3.54	**0.3922**	0.088	0.580	1.517	8.616
Deepseek R1	Standard	1.5b	DeepSeek AI	10.56	17.54	0.0697	0.018	0.206	11.730	50.824
Deepseek R1	Standard	8b	DeepSeek AI	0.19	8.72	0.2680	0.055	0.378	2.748	13.469
Deepseek R1	Instruct	8b	DeepSeek AI	0.51	7.26	0.2082	0.043	0.415	2.679	19.315
Deepseek R1	Distill	70b	DeepSeek AI	34.25	3.67	0.3295	0.088	0.502	1.799	11.505
GPT-4.1	Standard	-	OpenAI	**0.01**	**2.27**	0.3796	0.065	**0.594**	2.125	13.113

performs best ($s_{\text{cat}} = 0.3922$), while also maintaining low syntactic error and strong numerical accuracy. Among smaller models, Phi-3 Mini (3.8B) shows strong performance—especially in its quantized form—with minimal accuracy loss. In contrast, DeepSeek R1 (1.5B) and its 70B distilled variant exhibit high syntactic error, limiting their usability. Overall, instruction-tuned and mid-to-large models produce more accurate and consistent knowledge, though smaller models remain competitive in resource-constrained settings.

Effects of model size. To evaluate the impact of model size on extraction quality, we compared larger and smaller variants within the same model families: DeepSeek 1R (8B vs. 1.5B) and Qwen 2.5 (7B vs. 1B). We used the Wilcoxon signed-rank test to assess statistical significance across all dimensions. The Qwen2.5 models (1B and 7B) showed a statistically significant difference in syntax parsing success rates ($p < 0.001$), but no significant difference in semantic parsing ($p = 0.671$); however, both numerical ($p < 0.001$) and categorical ($p < 0.001$) prediction scores differed significantly, favoring the larger model. The Deepseek R1 models (1B and 8B) exhibited statistically significant differences across all evaluation metrics: syntax parsing ($p < 0.001$), semantic parsing ($p < 0.001$), numerical prediction ($p < 0.001$), and categorical prediction ($p < 0.001$), consistently favoring the larger 8B model. (For a complete overview see Figure 3).

Effects of model types. We compared different model variants to assess how instruction tuning and model type affect the quality of extracted commonsense knowledge. Specifically, we evaluated the LLaMA 3.1 models (Standard, Instruct, and Instant) at 8B parameters each, along with DeepSeek1R Standard and Instruct models, also at

8B parameters. A Friedman test revealed significant differences in both numerical ($\chi^2 = 27.3$, $p < 0.001$) and categorical ($\chi^2 = 9.37$, $p = 0.009$) scores across the three `LLaMA3-8B` variants. Chi-squared tests also showed significant differences in syntax ($\chi^2 = 597.2$, $p < 0.001$) and semantic ($\chi^2 = 77.7$, $p < 0.001$) parsing success rates. Pairwise comparisons revealed that `llama31_8b_standard` achieved the highest semantic parsing accuracy and consistently outperformed `llama31_8b_instant` in both numerical and categorical scores. Meanwhile, `llama31_8b_instruct` had the highest syntax parsing success but was outperformed by `standard` in numerical scores. The comparison between Deepseek R1 8B Instruct and Standard models revealed statistically significant differences across all metrics: syntax parsing (p < 0.001), semantic parsing (p < 0.001), numerical performance (p < 0.001), and categorical performance (p < 0.001), indicating consistent performance differences despite similar model sizes, with the Instruct model yielding better performance in numerical scores and the Standard model in categorical scores.

Effects of quantization. We examined the impact of quantization using the Phi-3 Mini model (3.8B parameters) in both 4-bit and 16-bit precision variants. The 4-bit quantized version (2.39 GB) is optimized for resource-constrained environments,[3] while the 16-bit version (7.64 GB) serves as a higher-precision baseline. To assess the performance difference, we conducted a Wilcoxon signed-rank test across all extracted dimensions, where they exhibited no significant difference in syntax parsing success rates (p = 0.317), but showed that the 16-bit precision variant showed significantly better performance in semantic parsing (p < 0.001), numerical prediction accuracy (Wilcoxon p = 0.000), and categorical prediction accuracy (Wilcoxon p = 0.000).

4.2.2 RQ2: Measurement Units. The measurement units of weight (kg, g, oz, lb), size (m, cm, mm, ft, in), and temperature ($^\circ K$, $^\circ F$ and $^\circ C$) were tested, where the mean average error (mae) was compared against the ground truth. As the Shapiro test showed a violation of the assumption of normality regarding the distribution of mae, the Friedman test was used as a non-parametric test. As it showed significant differences across all dimensions of measurement ($\chi^2(df = 2) = 1555.28, p < 0.000$ for temperature, $\chi^2(df = 4) = 467.38, p < 0.000$ for height, $\chi^2(df = 4) = 411.24, p < 0.000$ for length, $\chi^2(df = 4) = 726.41, p < 0.000$ for width, and $\chi^2(df = 3) = 288.50, p < 0.000$ for weight), Nemenyi post-hoc pair-wise analysis revealed that ounces perform significantly better than all the other measures, whereas kg, gram, and pounds showed no significant differences ($p > 0.11$). Predicting the width in inches significantly outperforms all other measures except centimeters; however, the height was best estimated in centimeters, and length in meters with centimeters being non-significantly worse. Furthermore, regarding temperature, Kelvin yields significantly more accurate predictions ($p < 0.000$), where Celsius performed significantly worse than the other two measurement units ($p < 0.000$).

4.2.3 RQ3: Context accuracy. Annotators were asked to indicate the dimensions for each concept that is not considered relevant. Majority voting was considered for these cases, which yielded in 34 context restrictions, most of them relating to temperature (87%), followed by size(13%). The majority of these restrictions align with common sense (i.e. the temperature of a bicycle, or a backpack were considered irrelevant). However, the mean relative error of predicted temperature values showed no significant variation across concepts, suggesting that contextual relevance cannot be reliably inferred from prediction variability alone. As an additional step, we considered the cases where the annotators could not reach a majority voting for the categorical concepts under the assumption that these indicate potentially irrelevant dimensions. We assessed whether the contextual dimension overlap between model outputs and human annotations exceeded chance expectations using a hypergeometric test. Several models showed statistically significant overlap at $p < 0.05$, including Qwen25 (7B) ($p < 0.0001$), DeepSeekR1 (1B) ($p = 0.0137$), Olmo2 ($p = 0.0192$), Llama4 ($p = 0.0311$), DeepSeekR1 (70B) ($p = 0.0221$), and Llama (30B) ($p = 0.0214$). Several other models—such as Llava, Llama3.1 (std), and DeepSeekR1 (7B)—showed near-significant overlap. In contrast, smaller or quantized models such as Phi3 mini demonstrated overlap not distinguishable from random baselines together with GPT4.1, and Llama3.1 (inst) ($p > 0.5$).

5 Discussion

We first examined how model type and size affect prediction quality (**RQ1**). As expected, instruction-tuned and larger models performed best overall, with LLama4 Scout and GPT-4.1 leading in categorical and numerical dimensions, respectively. Several open-source models (e.g., Qwen2.5 7B, Olmo2) also showed strong performance, while compact models like Phi-3 Mini performed surprisingly well—even in quantized form—highlighting their potential for use in robotics. Continued improvements in smaller models are likely to further close the gap regarding commonsense knowledge extraction. We also observed higher rates of syntax and semantic errors in models geared toward reasoning, such as DeepSeek R1. These errors likely stem from the models "overthinking" the prompt—producing overly verbose outputs or exceeding the maximum allowed output tokens (500 in our case), which resulted in incomplete or malformed responses. When possible, it would be beneficial to use these models with reasoning disabled to ensure more concise and parseable responses.

To answer **RQ2**, we examined how measurement unit selection affects prediction quality. While several differences were statistically significant, none indicated major performance issues. The results suggest that LLMs handle metric units (e.g., kilograms, centimeters, Kelvin) more consistently, likely due to their frequency in training data. Overall, careful unit selection can enhance reliability but is rarely a major bottleneck.

RQ3 explored how contextually relevant dimensions can be determined from model outputs. While annotators identified temperature as often irrelevant, model variance across concepts was insufficient to reflect this contextual irrelevance. However, a hypergeometric test revealed significant alignment between model-predicted relevant categorical dimensions and human judgments for several models (e.g., Qwen2.5 7B, DeepSeekR1 1B, DeepSeekR1 70b). These results suggest that, although reasoning-oriented models do not necessarily excel at predicting categorical or numerical values,

[3] https://huggingface.co/microsoft/Phi-3-mini-4k-instruct-gguf

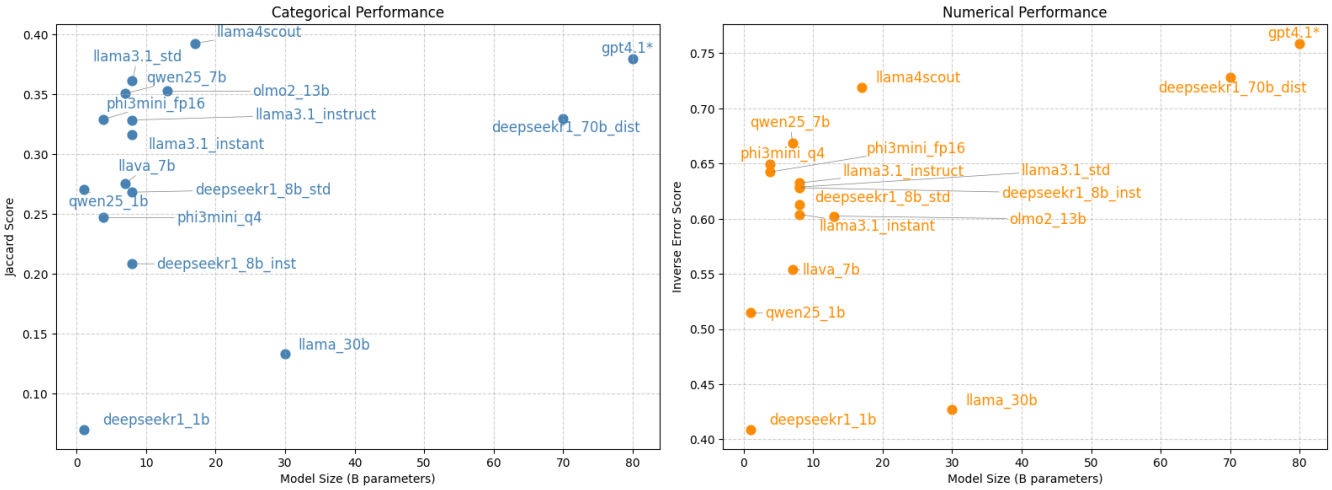

Figure 3: The model performances visualized in terms of the model sizes. The size of GPT4.1 is only representative.

they may be better suited for identifying contextually relevant dimensions—i.e., estimating the context vector—where interpretive reasoning is more valuable than precision.

Limitations. This study has several limitations. First, the evaluation is restricted to a limited set of 80 common object concepts, which may not reflect the variability present in more specialized or less frequently encountered categories. Second, we focus on a predefined set of quality dimensions, which, while grounded in theory and robotics relevance, do not capture the full range of potentially meaningful attributes. While we provided the concept description, which might not always be readily available, techniques such as generated knowledge prompting [22] could be used to create the contextual information and overcome this limitation. Lastly, the ground truth was established using a limited pool of human annotators, which may introduce bias or noise in both the value estimates and relevance judgments. Additionally, while we include quantized models to reflect resource-constrained deployment scenarios, quantization effects were only explored using a single model configuration, limiting generalizability across architectures.

Future Work. While this study primarily introduced and evaluated the knowledge acquisition pipeline, future work will integrate the resulting knowledge graphs into robotic systems and use the extracted commonsense knowledge to auto-generate planning domains. To assess generalizability beyond common objects, upcoming experiments will target less typical and domain-specific items from the 'long-tail' of concepts. We also aim to couple model selection with ontology-based reasoning for task-aware model choice. In parallel, research on conceptual spaces will refine the weighting of quality dimensions. Finally, the extracted prototypical representations may serve as a valuable resource for researchers studying how LLMs encode conceptual knowledge, offering insights into their alignment with cognitive models of categorization and feature salience.

6 Conclusion

This paper introduced a pipeline for extracting structured commonsense knowledge from Large Language Models and integrating it into a symbolic knowledge graph grounded in an ontology designed to accommodate robotic knowledge acquisition [1]. Grounded in the theory of conceptual spaces, we focused on object properties relevant to embodied agents, covering both categorical (e.g., material, color, location) and quantifiable (e.g., weight, size, temperature) quality dimensions. Our evaluation across a set of different LLMs shows that even smaller and quantized models can yield outputs closely aligned with human annotations—especially for core physical dimensions—while performance varies by model type, unit selection, and contextual relevance. Reasoning-focused models, though less accurate in prediction tasks, show potential for identifying contextually relevant features.

By providing structured representation and grounding in a robotic ontology, this work strengthens the bridge between LLM-driven commonsense knowledge and the demands of symbolic reasoning in safety-critical systems. The resulting knowledge graphs can support affordance inference, task planning, and other downstream reasoning tasks while also providing a testbed for analyzing how LLMs encode conceptual and prototypical knowledge. This positions our approach as both a practical contribution to robotic knowledge engineering and a tool for potential future interdisciplinary research on the cognitive modeling capabilities of LLMs.

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